The feasible region Ω

The mathematical formulation has several mistakes:

- To apply the Brower theorem, Ω has to be convexe.

- f_0 has to be continuous and invariant. That is, $f_0(\omega) \in \Omega$, for all $\omega \in \Omega$. This fact can be imposed by force, but perhaps it would not correspond to the fact that one wishes to model.

- Even if $f_0 : \Omega \to \Omega$ is continuous and invariant, the values of $\mathbf{x}(t+1)$ need not belong to Ω, even though $\mathbf{x}(t) \in \Omega$. This is due to the noise $\sigma \cdot \eta(t)$.